DecodeLabs | Batch 2026

Data Analytics — Project 1

Data Cleaning & Preparation

Dataset: 1,200 e-commerce orders | Tool: Python / pandas

1. Project Overview

The goal of Project 1 is to transform a raw, unstructured dataset into a clean and reliable source of truth. Before any analysis or modeling can be done, the data must be verified for integrity. This report documents every step taken to clean the dataset.

The dataset contains 1,200 order records across 14 columns covering order IDs, dates, products, prices, payment methods, and customer referral sources.

2. Cleaning Steps

Step 1 — Missing Values

The **CouponCode** column had **309 missing values**. Rather than deleting those rows (which would reduce statistical power), the nulls were filled with the string **NO_COUPON** — logically meaning the order was placed without a coupon code. This is called Strategic Imputation.

Column affected	**CouponCode**
Missing values found	**309**
Strategy used	**fillna('NO_COUPON')**
Missing values remaining	**0**

Step 2 — Duplicate Records

A check was performed for fully duplicated rows and for repeated OrderIDs. No duplicates were found in either case, confirming the dataset has unique records.

Duplicate rows	**0**
Duplicate OrderIDs	**0**
Action taken	**None required**

Step 3 — Date Format Standardisation

All dates were converted to the ISO 8601 standard format **YYYY-MM-DD** using pandas *to_datetime*. Zero parsing errors were encountered, confirming all dates were valid.

Target format	**YYYY-MM-DD (ISO 8601)**
Date range	**2023-01-04 → 2025-12-27**

Invalid dates found	**0**

Step 4 — Text Standardisation

All text columns were cleaned in two ways: leading and trailing whitespace was removed using *str.strip()*, and Proper Case was applied using *str.title()* on columns where it made sense. ID and code columns (OrderID, CouponCode, TrackingNumber) were left as-is.

Columns cleaned	**Product, OrderStatus, PaymentMethod, ReferralSource, ShippingAddress**
Whitespace removed	**str.strip()**
Case applied	**str.title() (Proper Case)**

Step 5 — Numeric Precision

UnitPrice and TotalPrice were rounded to two decimal places to ensure consistent currency formatting across all records.

UnitPrice	**Rounded to 2 decimal places**
TotalPrice	**Rounded to 2 decimal places**

Step 6 — TotalPrice Integrity Check

A verification was performed to confirm that TotalPrice equals Quantity × UnitPrice for every row. Zero mismatches were found, confirming the financial data is internally consistent.

Formula checked	**TotalPrice = Quantity × UnitPrice**
Mismatches found	**0**
Result	**All values are consistent**

3. Final Dataset State

Total rows	**1,200**
Total columns	**14**

Missing values	**0**
Duplicate rows	**0**
Duplicate OrderIDs	**0**
Date format errors	**0**
Output file	**Dataset_Cleaned.xlsx**

4. Conclusion

The dataset has been fully cleaned and is ready for exploratory analysis. All six cleaning steps passed with zero remaining issues. The cleaned file has been saved as **Dataset_Cleaned.xlsx** and serves as the single source of truth for all subsequent projects.